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Inventor(s)	Shetty; Shivananda et al.

Memory location mapping and unmapping

Abstract

A method of operating a memory circuit is disclosed. The memory circuit includes a memory array having a memory portion and a spare portion. The method includes receiving a first write command to a first memory address, where the first memory address has a status of being mapped to a first spare memory address, and where the first memory address corresponds to a first memory location in the memory portion and the first spare memory address corresponds to a first spare memory location in the spare portion. The method also includes performing, in response to the first write command, a first write operation by attempting to write first data to the first memory location, determining if the first write operation is successful, and unmapping, in response to the first write operation of being successful, the first memory address from the first spare memory address.

Inventors: **Shetty; Shivananda (Fremont, CA), Amato; Stefano (San Jose, CA)**

Applicant: **Infineon Technologies LLC** (San Jose, CA)

Family ID: **95339561**

Assignee: **Infineon Technologies LLC (San Jose, CA)**

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Primary Examiner: Dinh; Son T

Background/Summary

TECHNICAL FIELD

(1) The present invention relates generally to an electronic system and method, and, in particular embodiments, to a system and method for repairing and releasing memory locations in a memory array.

BACKGROUND

(2) Programmable memories are useful in many applications. There are many different types of programmable memories including, but not limited to, resistive random access memory (RRAM), magnetoresistive random access memory (MRAM), phase change memory (PCM), ferroelectric random access memory (FERAM), programmable read only memory (PROM), electrically erasable ROM (EEPROM) and Flash memory.

(3) In order to improve reliability of programmable memories, in addition to a main memory array area, a spare memory area is sometimes included. If a particular memory cell location of the main memory array area is found to be faulty, the particular memory cell location may be mapped or in instances remapped to a location in the spare memory area. In subsequent write or read operations, data would be written to or read out from the mapped spare memory location. The memory cell location that is considered faulty would not be utilized to store data again. Current techniques require large amounts of spare memory area to meet reliability requirements, and are, therefore,

expensive.

SUMMARY

(4) One aspect of an embodiment is a method of operating a memory circuit including a memory array having a memory portion and a spare portion. The method includes receiving a first write command to a first memory address, where the first memory address has a status of being mapped to a first spare memory address, and where the first memory address corresponds to a first memory location in the memory portion and the first spare memory address corresponds to a first spare memory location in the spare portion. The method also includes performing, in response to the first write command, a first write operation by attempting to write first data to the first memory location, determining if the first write operation is successful, and unmapping, in response to the first write operation of being successful, the first memory address from the first spare memory address.

(5) Another aspect of an embodiment is a memory circuit including a memory array including a memory portion and a spare portion, and a controller, configured to perform a first write operation by attempting to write first data to a first memory location in the memory portion, where a first memory address corresponds with the first memory location, and the first memory address has a status of being mapped to a first spare memory address corresponding with a first spare memory location in the spare portion when performing the first write operation, and in response to a determination that the first data was successfully written to the first memory location during the first write operation, unmap the first memory address from the first spare memory address.

(6) Another aspect of an embodiment is a system including an external circuit configured to generate instructions, and a memory circuit configured to receive the instructions from the external circuit, the memory circuit including a memory array including a memory portion and a spare portion, and a controller, configured to, in response to a first instruction from the external circuit perform a first write operation by attempting to write first data to a first memory location in the memory portion, where a first memory address corresponds with the first memory location, and the first memory address has a status of being mapped to a first spare memory address corresponding with a first spare memory location in the spare portion when performing the first write operation, and in response to a determination that the first data was successfully written to the first memory location during the first write operation, unmap the first memory address from the first spare memory address.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of the present invention, and the advantages thereof, reference is now made to the following description taken in conjunction with the accompanying drawings, in which:

(2) FIG. **1** illustrates a block diagram of an embodiment of a memory system according to some embodiments of the present invention;

(3) FIGS. **2**A and **2**B illustrate graphical conceptualizations of methods performed on a portion of a memory according to some embodiments of the present invention.

(4) FIG. **3** illustrates a graphical conceptualization of methods according to some embodiments of the present invention;

(5) FIG. **4** illustrates a flowchart diagram of a method of using a memory according to some embodiments of the present invention; and

(6) FIG. **5** illustrates a flowchart diagram of a method of using a memory according to some embodiments of the present invention.

(7) FIG. **6** illustrates a flowchart diagram of a method of using a memory according to some embodiments of the present invention.

(8) FIG. **7** illustrates a processing system that can be used to implement portions of embodiment systems.

(9) Corresponding numerals and symbols in different figures generally refer to corresponding parts unless otherwise indicated. The figures are drawn to clearly illustrate the relevant aspects of the preferred embodiments and are not necessarily drawn to scale. To more clearly illustrate certain embodiments, a letter indicating variations of the same structure, material, or process step may follow a figure number.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(10) The making and using of the presently preferred embodiments are discussed in detail below. It should be appreciated, however, that the present invention provides many applicable inventive concepts that can be embodied in a wide variety of specific contexts. The specific embodiments discussed are merely illustrative of specific ways to make and use the invention, and do not limit the scope of the invention.

(11) Some memory technologies are susceptible to errors which cause write operations to fail to change the state of a memory location. In some memory technologies, the failures may be intermittent, such that subsequent successful writing operations may be performed at memory locations having previous failed writing operations. Memory technologies which are susceptible to intermittent errors include, but are not limited to, resistive random access memory (RRAM), magnetoresistive random access memory (MRAM), phase change memory (PCM), ferroelectric random access memory (FERAM), programmable read only memory (PROM), electrically erasable ROM (EEPROM) and Flash memory

(12) For example, some memories use filament technologies to conditionally establish a low resistance conductive path between addressable electrodes of a memory cell, where the logic state of the memory cell is encoded in the conductivity state. When the connection and disconnection of the filament is properly induced by the writing control signals, the writing operations are successful. And conversely, when the connection and disconnection of the filament is not properly induced by the writing control signals, the writing operations are not successful.

(13) When writing to such a filament based memory or other types of memory, sometimes a write operation fails to properly change the conductivity state or other corresponding parameter(s) of the memory cell, for example, as determined by a subsequent read operation to determine whether the write operation was successful. In some embodiments, when a write operation to a first memory location is not successful, the data to be stored in the first memory location is written to another memory location, for example, in a physical spare memory area. In one embodiment, the first memory location may be considered or labeled as failed memory bits or failed bits. In some embodiments, failed bits may be permanently discarded and never be used to store data.

(14) In addition, the address of the first memory location is mapped to the address of the physical spare memory location to which the data was written, or in embodiments considered failed bits. For example, the first memory address may be stored in a mapping lookup table associating the first memory address and the address of the physical spare memory location to which the data was written. Consequently, subsequent read operations attempting to read data from the first memory address, after referencing the mapping lookup table, identify the address of the spare memory location, and fetch and return the data stored in the physical spare memory location to which the data was written.

(15) In some memory technologies, the failures may be intermittent, such that subsequent successful writing operations may be performed at memory locations having previous failed writing operations or failed bits. Advantageously, various embodiments discussed herein reuse memory locations which have previously failed writing operations or been labeled failed bits. Accordingly, memories having features discussed herein have improved reliability. Some memories having features discussed herein may use less area because less spare memory may be used.

(16) For example, after a particular memory location has failed a write operation, subsequent read

operations access the spare memory location to which the memory location has been mapped. However, subsequent write operations may be performed on the memory location. As a result, the memory location is once again available for writing operations. Consequently, the memory location is not forever discarded, and reliability of the memory is improved and/or less physical spare memory is used.
(17) FIG. **1** illustrates an embodiment memory system that includes memory **100**. As shown, memory **100** includes memory array **104**, row decoder **102**, word line drivers **103**, controller **106**, sense amplifiers **112**, write drivers **113**, column decoder **110**, I/O logic **114** coupled to an external data bus, and reference current generator **108**. In another embodiment, memory **100** may include more than one memory arrays **104** (not shown). Reference current generator **108** may also be referred to as an adjustable current source. In some embodiments, memory **100** may be coupled to an external circuit **120**, for example, as part of a computer system.
(18) In various embodiments, the components of memory **100** may be disposed on a single monolithic semiconductor integrated circuit, such as a single semiconductor substrate, and/or on the same monolithic semiconductor integrated circuit as other disclosed system components.
(19) As shown, memory array **104** includes a memory array of memory cells **105** and a spare area of memory cells **107**. In one embodiment, memory array of memory cells **105** and a spare area of memory cells **107** may be disposed physically adjacent to one another. In other embodiments, they may be disposed physically separated from one another, or in different memory arrays **104**. Each memory cell **105** is connected to a respective word line (WL1 . . . WLn) and to a respective bit line (BL1 . . . BLm). Similarly, each memory cell **107** is connected to a respective word line (WL1 . . . WLn) and to a respective spare bit line (SBL1 . . . SBLn). Memory array **104** in FIG. **1** illustrates that memory array of memory cells **105** and spare area of memory cells **107** are disposed next to one another and may share word lines. This example is merely one of many embodiments of the present disclosure. In other embodiments, a spare area may be disposed in other physical locations and may not share any word lines or bit lines with memory array memory cells **105**. In other embodiments, a spare area may share bit lines with memory area cells **105**.
(20) During normal read operations of the memory, row decoder **102** decodes a row address based on address data ADDR provided at the input to I/O logic **114**. The decoded row address is used to select one of word lines (WL1 . . . WLn) via word line drivers **103**, which activates a row of memory cells **105** that generate respective read currents on respective bit lines (BL1 . . . BLm and SBL1 . . . SBLn). The current of each bit line is compared to a reference current produced by reference current generator **108** using sense amplifiers **112**. Column decoder **110** selects columns of memory array **104** to be routed to sense amplifiers **112**, the output of which is routed to data lines DATA via I/O logic **114**. Controller **106** is a memory controller that controls the operation of the memory. In some embodiments, reference current generator **108** may be implemented using a current digital-to-analog converter (CDAC).
(21) During normal write operations of the memory, row decoder **102** decodes a row address based on address data ADDR provided at the input to I/O logic **114**. The decoded row address is used to select one of word lines (WL1 . . . WLn) via word line drivers **103**, which activates a row of memory cells **105** and **107** that receive respective write data on respective bit lines (BL1 . . . BLm and SBL1 . . . SBLn) from write drivers **113** according to data provided at the input to I/O logic **114**. Those activated memory cells **105** and **107** connected to the selected word line are written with the write data.
(22) In some embodiments, after a write operation or after every write operation, the controller **106** causes a verification read operation to be performed to verify that the write operation was successful. In some embodiments, the verification read operation includes performing a read operation, and if the read operation retrieves data matching that which was written, the write operation is determined to be successful. In some embodiments, if the read operation fails to retrieve data matching that which was written, the write operation is determined to be unsuccessful.

In some embodiments, as part of the verification read operation, more than one write and verification read is performed before determining that the write operation is unsuccessful. In one embodiment, more than one write attempt and each followed by a verification read will be performed. If it is verified that all write attempts failed to write the data correctly, the verification read operation may conclude the write operation as a failure or not successful.

(23) In some embodiments, if the controller **106** determines that a write operation to a first memory location was not successful, the controller **106** causes the data that was to be stored in the first memory location to be written to another memory location, for example, in the physical spare memory area.

(24) In addition, the controller **106** may map the address of the first memory location to the address of the physical spare memory location to which the data was written. For example, the controller **106** may store the first memory address in a mapping lookup table associating the first memory address and the address of the physical spare memory location to which the data was written. Consequently, the controller **106** may reference the mapping lookup table as part of performing subsequent read operations, and, after finding that the first memory address is to be read, may cause the read operations to, instead of reading data from the first memory location of the memory array, may retrieve the address of the spare memory location from the mapping lookup table, and fetch and return the data stored in the physical spare memory location to which the data was written.

(25) However, in some embodiments, the controller **106** causes subsequent write operations addressed to the memory location to be performed on the memory location, for example, even if the memory location is currently mapped. Accordingly, despite having previous failed write operations, the memory location is available for subsequent write operations. Consequently, the memory location is not discarded.

(26) FIGS. **2**A and **2**B illustrate graphical conceptualizations of methods performed on a portion of a memory **200** according to some embodiments of the present invention. FIGS. **2**A and **2**B show a memory array area **210** and a spare memory area **220** of memory **200** in different mapping states.

(27) FIG. **2**A shows the memory array area **210** having memory location **215** mapped to spare memory location **225** in the spare memory area **220**, for example, as a consequence of memory location **215** having failed one or more write operation attempts, depending on the system requirements.

(28) For example, as discussed above, the data which was to have been written to memory location **215** during the failed write operation attempt may have been written, instead, to spare memory location **225**. In addition, memory location **215** may have been mapped to spare memory location **225**, for example, using a mapping lookup table, as discussed above, for example.

(29) FIG. **2**B shows the memory array area **210** having memory location **215** unmapped or released from spare memory location **225** in the spare memory area **220**, for example, as a consequence of memory location **215** having subsequently been successfully written to.

(30) For example, as discussed above, a subsequent write attempt may have successfully written data to memory location **215**. In addition, spare memory location **225** may be unmapped from memory location **215**, for example, by removing the entry in the mapping lookup table associating spare memory location **225** with memory location **215**.

(31) FIG. **3** illustrates a graphical conceptualization of methods according to some embodiments of the present invention. FIG. **3** shows a memory array location **310** and a spare memory location **320** during different operational phases **350**, **360** and **370**. In embodiments, each memory array location **310** and spare memory location **320** may include one or more memory cells, such as memory array memory cells **105** and spare area memory cells **107**, as best shown in FIG. **1**.

(32) During phase **350**, the memory attempts to write data to memory array location **310**, but the attempt fails. In response to the failed attempt, the memory subsequently writes the data to spare memory location **320**. An association of memory locations between memory array location **310** and

spare memory location **320** is established, for example via address mapping or remapping. In one embodiment, a subsequent read request of data stored in memory array location **310** will cause data stored in spare memory location **320** to be read out.
(33) Next, during phase **360**, the memory attempts to write new data to memory array location **310**, but the attempt fails. In response to the failed attempt, the memory subsequently writes the new data to spare memory location **320**, which is already associated with memory array location **310** via previous mapping in phase **350**.
(34) During phase **370**, after phase **360**, the memory attempts to write next data to memory array location **310**, and the attempt succeeds. In response to the successful attempt, the memory subsequently disassociates or unmaps spare memory location **320** from memory array location **310**. Accordingly, because of being unmapped, spare memory location **320** is available to be used as a spare if a subsequent attempt to write data to memory array location **310** or another memory location fails. In one embodiment, any subsequent read request of data stored in memory array location **310** will no longer be read out from spare memory location **320**.
(35) FIG. **4** illustrates a flowchart diagram of a method **400** of using a memory according to some embodiments of the present invention. Method **400** may be performed, for example, by a memory, such as memory **100**.
(36) At step **405**, the controller of the memory receives an address for data to be written to. For example, the address may be part of a data communication received from an external circuit.
(37) At step **415**, the controller of the memory receives data for the write operation. For example, the data may be part of the data communication received from the external circuit which also included the address the data is to be written to. In some embodiments, the data communication also indicates that the operation to be performed is a write operation.
(38) At step **425**, the controller of the memory causes a write operation to be performed which attempts to write the data received at step **415** to the memory location corresponding with the address received at step **405**. In some embodiments, the controller causes the write operation to be performed even if previous attempts to write data to the memory location corresponding with the address received at step **405** have failed and the address received at step **405** is currently mapped. For example, in some embodiments, the controller does not access a mapping lookup table to determine whether the address received at step **405** is in the mapping lookup table or is associated with another memory location in the mapping lookup table as part of the write operation or as a condition for the write operation.
(39) At step **435**, the controller determines whether the write operation performed at step **425** was successful. In some embodiments, the controller causes a verification read operation to be performed to access the data stored at the memory location corresponding with the address received at step **405** to determine whether the data stored at the memory location corresponding with the address received at step **405** matches the data received at step **415** and which was the subject of the write operation at step **425**. In some embodiments, the write operation may be determined as not successful after one or more than one consecutive failed write attempts and verification read operations.
(40) If, at **435**, the controller determines that the write operation was not successful, at **445**, the controller of the memory causes a write operation to be performed which attempts to write the data received at step **415** to a memory location corresponding with a second memory address, such as a memory address for a memory located in a spare area of the memory.
(41) At step **455**, the controller determines whether the write operation performed at step **445** was successful. In some embodiments, the controller causes a read operation to be performed to access the data stored using the second memory address to determine whether the data stored with the address received at step **405** matches the data received at **415** and which was the subject of the write operation of step **445**. In some embodiments, the write operation may be determined as not successful after one or more than one consecutive failed verification read operations.

(42) If the controller determines, at step **455**, that the write operation was not successful, at step **445**, the controller of the memory causes a write operation to be performed which attempts to write the data received at step **415** to another memory location, such as a memory location in a spare area of the memory.
(43) If the controller determines, at step **455**, that the write operation was successful, at step **465**, the controller of the memory causes the address received at step **405** to be mapped to the other memory address to which the data received at step **415** was successfully written. For example, the address received at step **405** may be stored in a mapping lookup table associating the address received at step **405** and the address of the other memory address to which the data received at step **415** was successfully written.
(44) If, at step **435**, the controller determines that the write operation was successful, at step **475**, the controller of the memory determines whether the address received at step **405** is mapped to another memory address. For example, the controller may access a mapping lookup table to determine whether the address received at step **405** is in the mapping lookup table or is associated with another memory location in the mapping lookup table.
(45) If, at step **475**, the controller determines that the address received at step **405** is not in the mapping lookup table nor associated with another memory location in the mapping lookup table, at step **485**, the controller maintains the current mapping of the address received at step **405**. For example, the controller may refrain from writing the address received at step **405** in the mapping lookup table.
(46) If, at step **475**, the controller determines that the address received at step **405** is in the mapping lookup table or is associated with another memory location in the mapping lookup table, at step **495**, the controller maintains the original mapping of the address received at step **405**. For example, the controller may unmap or disassociate the current mapping of the address received from the address received at step **405**. For example, the controller may remove an entry in the mapping lookup table associating the other memory location with the memory location received at step **405**.
(47) FIG. **5** illustrates a flowchart diagram of a method **500** of using a memory according to some embodiments of the present invention. Method **500** may be performed, for example, by a memory, such as memory **100**.
(48) At step **505**, the controller of the memory receives an address for data to be written to. For example, the address may be part of a data communication received from an external circuit.
(49) At step **515**, the controller of the memory receives data for the write operation. For example, the data may be part of the data communication received from the external circuit which also included the address the data is to be written to. In some embodiments, the data communication also indicates that the operation to be performed is a write operation.
(50) At step **525**, the controller of the memory causes a write operation to be performed which attempts to write the data received at step **515** to the memory location corresponding with the address received at step **505**. In some embodiments, the controller causes the write operation to be performed even if previous attempts to write data with the address received at step **505** have failed (the memory location corresponds with the address labeled as “failed bits”) and the address received at step **505** is currently mapped. For example, in some embodiments, the controller does not access a mapping lookup table to determine whether the address received at step **505** is in the mapping lookup table or is associated with another memory location in the mapping lookup table as part of the write operation or as a condition for the write operation.
(51) At step **535**, the controller determines whether the write operation performed at step **525** was successful. In some embodiments, the controller causes a verification read operation to be performed with the address received at step **505** to determine whether the data stored at the address received at step **505** matches the data received at step **515** and which was the subject of the write operation at step **525**.
(52) If, at step **535**, the controller determines that the write operation was successful, at step **545**,

the controller of the memory may optionally cause one or more write attempts and read operations to be performed, for example, in response to one or more read instructions requesting data from the memory location corresponding with the address received at step **505**.

(53) At step **555**, the controller of the memory may optionally cause one or more read or write operations to be performed, for example, in response to one or more read or write instructions requesting data from one or more addresses different from the address received at step **505** or requesting data to be written to one or more addresses different from the address received at step **505**.

(54) If, at step **535**, the controller determines that the write operation was not successful, at step **565**, the controller of the memory causes a write operation to be performed which attempts to write the data received at step **515** to a location corresponding with a second memory address, such as a memory location in a spare area of the memory.

(55) At step **575**, the controller determines whether the write operation performed at step **565** was successful. In some embodiments, the controller causes a read operation to be performed to determine whether the data stored at the address received at step **505** matches the data received at **515** and which was the subject of the write operation at step **565**.

(56) If the controller determines, at step **575**, that the write operation was not successful, at step **565**, the controller of the memory causes a write operation to be performed which attempts to write the data received at step **515** to a memory location corresponding with another memory address, such as a memory address for a spare area of the memory.

(57) If the controller determines, at step **575**, that the write operation was successful, at step **585**, the controller of the memory causes the address received at step **505** to be mapped to the other memory address to which the data received at step **515** was successfully written. For example, the address received at step **505** may be stored in a mapping lookup table associating the address received at step **505** and the address of the other memory address to which the data received at step **515** was successfully written.

(58) At step **595**, the controller of the memory may optionally cause one or more read operations to be performed, for example, in response to one or more read instructions requesting data from the memory location corresponding with the address received at step **505**.

(59) At step **597**, the controller of the memory may optionally cause one or more read or write operations to be performed, for example, in response to one or more read or write instructions requesting data from one or more addresses different from the address received at step **505** or requesting data to be written to one or more addresses different from the address received at step **505**.

(60) FIG. **6** illustrates a flowchart diagram of a method **600** of using a memory according to some embodiments of the present invention. Method **600** may be performed, for example, by a memory, such as memory **100**.

(61) At step **605**, the controller of the memory receives an address for an instruction. The instruction may be, for example, for a read operation which reads data from the address or the instruction may be, for example, for a write operation which to write data to the address. The address may be, for example, part of a data communication received from an external circuit.

(62) At step **615**, the controller of the memory receives data identifying the instruction. For example, the controller may receive data indicating that the instruction is for a read operation which reads data using the address. Alternatively, the controller may receive data indicating that the instruction is for a write operation which to write data to the address. Accordingly, the controller may also receive the data to be written for the write instruction. The data may be, for example, part of the data communication received from the external circuit which also includes the address received at **605**.

(63) At step **620**, the controller determines that the instruction received at step **615** is for a write operation or determines that the instruction received at **615** is for a read operation.

(64) If at step **620** the controller determines that the instruction received at step **615** is for a write operation, at step **625**, the controller of the memory causes a write operation to be performed which attempts to write the data received at step **615** to the memory location corresponding with the address received at step **605**. In some embodiments, the controller causes the write operation to be performed even if previous attempts to write data to the memory location corresponding with the address received at step **605** have failed and the address received at step **405** is currently mapped. For example, in some embodiments, the controller does not access a mapping lookup table to determine whether the address received at step **605** is in the mapping lookup table or is associated with another memory location in the mapping lookup table as part of the write operation or as a condition for the write operation.
(65) In some embodiments, at step **625**, the control also performs other actions described above, for example, with reference to methods **400** and **500**. For example, the controller may determine whether the write operation was successful, and perform further actions to implement mapping and unmapping techniques according to principles and aspects described elsewhere herein.
(66) If at step **620** the controller determines that the instruction received at step **615** is for a read operation, at step **635**, the controller of the memory determines whether the address received at step **605** is mapped to an alternative memory location.
(67) If at step **635** the controller of the memory determines that the address received at step **605** is mapped to an alternative memory location, the controller causes data to be read from the alternative memory location, and returns the fetched data as a response to the read instruction received at step **615**.
(68) If at step **635** the controller of the memory determines that the address received at step **605** is not mapped to an alternative memory location, the controller causes data to be read from the memory location corresponding with the address received at step **605**, and returns the fetched data as a response to the read instruction received at step **615**.
(69) FIG. **7** illustrates a processing system that can be used to implement portions of embodiment systems.
(70) Referring now to FIG. **7**, a block diagram of a processing system **700** is provided in accordance with an embodiment of the present invention. The processing system **700** depicts a general-purpose platform and the general components and functionality that may use embodiments described herein, such as memory **100**. For example, processing system **700** may be used to implement some or all of the processing used to implement all or portions of the embodiment methods described herein.
(71) Processing system **700** may include, for example, a central processing unit (CPU) **702**, and memory **704** connected to a bus **708**. Memory **704** may be or include an embodiment of memory **100**. Processing system **700** may be configured to perform the processes discussed above according to programmed instructions stored in memory **704** or on other non-transitory computer readable media. The processing system **700** may further include, if desired or needed, a display adapter **710** to provide connectivity to a local display **712** and an input-output (I/O) adapter **714** to provide an input/output interface for one or more input/output devices **716**, such as a mouse, a keyboard, flash drive or the like.
(72) The processing system **700** may also include a network interface **718**, which may be implemented using a network adaptor configured to be coupled to a wired link, such as a network cable, USB interface, or the like, and/or a wireless/cellular link for communications with a network **720**. The network interface **718** may also comprise a suitable receiver and transmitter for wireless communications. It should be noted that the processing system **700** may include other components. For example, the processing system **700** may include hardware components power supplies, cables, a motherboard, removable storage media, cases, and the like if implemented externally. These other components, although not shown, may be considered to be part of the processing system **700**. In some embodiments, processing system **700** may be implemented on a single monolithic

semiconductor integrated circuit and/or on the same monolithic semiconductor integrated circuit as other disclosed system components.
(73) Embodiments of the present invention are summarized here. Other embodiments can also be understood from the entirety of the specification and the claims filed herein.
(74) Example 1. A method of operating a memory circuit, the memory circuit including a memory array having a memory portion and a spare portion, the method including: receiving a first write command to a first memory address, where the first memory address has a status of being mapped to a first spare memory address, and where the first memory address corresponds to a first memory location in the memory portion and the first spare memory address corresponds to a first spare memory location in the spare portion; performing, in response to the first write command, a first write operation by attempting to write first data to the first memory location; determining if the first write operation is successful; and unmapping, in response to the first write operation of being successful, the first memory address from the first spare memory address.
(75) Example 2. The method of example 1, where the first memory address having the status of being mapped to the first spare memory address indicates that the first memory location is considered failed memory bits.
(76) Example 3. The method of one of examples 1 or 2, subsequent to the first spare memory address being unmapped from the first memory address, further including:
(77) Example 4. The method of one of examples 1 to 3, where unmapping the first memory address from the first spare memory address includes disassociating the first memory address from the first spare memory address in a lookup table.
(78) Example 5. The method of one of examples 1 to 4, further including: performing a second write operation by attempting to write second data to the first memory location when the first memory address does not have a status of being mapped to a spare memory address; and in response to a determination that the second data was not successfully written to the first memory location: mapping the first memory address to a second spare memory address, where the second spare memory address is not mapped to any memory address; and writing the second data to a second spare memory location in the spare portion, the second spare memory location corresponding with the second spare memory address.
(79) Example 6. The method of one of examples 1 to 5, where the first spare memory location is the same as the second spare memory location.
(80) Example 7. The method of one of examples 1 to 6, where mapping the first memory address to the second spare memory address includes associating the first memory address with the second spare memory address in a lookup table.
(81) Example 8. The method of one of examples 1 to 7, further including, before performing the first write operation, reading data for the first memory address from the first spare memory location.
(82) Example 9. The method of one of examples 1 to 8, where determining if the first write operation is successful further including:
(83) Example 10. The method of one of examples 1 to 9, where the memory portion is physically separated from the spare portion in the memory array.
(84) Example 11. A memory circuit, including: a memory array including a memory portion and a spare portion; and a controller, configured to: perform a first write operation by attempting to write first data to a first memory location in the memory portion, where a first memory address corresponds with the first memory location, and the first memory address has a status of being mapped to a first spare memory address corresponding with a first spare memory location in the spare portion when performing the first write operation; and in response to a determination that the first data was successfully written to the first memory location during the first write operation, unmap the first memory address from the first spare memory address.
(85) Example 12. The memory circuit of example 11, where the memory portion of the memory

array and the spare portion of the memory array include resistive random access memory (RRAM) cells.
(86) Example 13. The memory circuit of one of examples 11 or 12, where the controller is configured to unmap the first memory address from the first spare memory address by disassociating the first memory address from the first spare memory address in a lookup table.
(87) Example 14. The memory circuit of one of examples 11 to 13, where the controller is configured to: perform a second write operation by attempting to write second data to the first memory location when the first memory address does not have a status of being mapped to a spare memory address; and in response to a determination that the second data was not successfully written to the first memory location: write the second data to a second spare memory location in the spare portion, the second spare memory location corresponding with a second spare memory address; and map the first memory address to the second spare memory address.
(88) Example 15. The memory circuit of example 14, where the controller is configured to map the first memory address to the second spare memory address by associating the first memory address with the second spare memory address in a lookup table.
(89) Example 16. The memory circuit of one of examples 11 to 15, where the controller is configured to, before the first write operation is performed, read data for the first memory address from the first spare memory location.
(90) Example 17. The memory circuit of example 16, where the controller is configured to, after the first memory address is unmapped from the first spare memory address, read data for the first memory address from the first memory location.
(91) Example 18. A system, including: an external circuit configured to generate instructions; and a memory circuit configured to receive the instructions from the external circuit, the memory circuit including: a memory array including a memory portion and a spare portion; and a controller, configured to, in response to a first instruction from the external circuit: perform a first write operation by attempting to write first data to a first memory location in the memory portion, where a first memory address corresponds with the first memory location, and the first memory address has a status of being mapped to a first spare memory address corresponding with a first spare memory location in the spare portion when performing the first write operation; and in response to a determination that the first data was successfully written to the first memory location during the first write operation, unmap the first memory address from the first spare memory address.
(92) Example 19. The system of example 18, where the controller is configured to unmap the first memory address from the first spare memory address by disassociating the first memory address from the first spare memory address in a lookup table.
(93) Example 20. The system of one of examples 18 or 19, where the controller is configured to, in response to a second instruction from the external circuit:
(94) Example 21. The system of one of examples 18 to 20, where the controller is configured to: before the first write operation is performed, read data for the first memory address from the first spare memory location; and after the first memory address is unmapped from the first spare memory address, read data for the first memory address from the first memory location.
(95) While this invention has been described with reference to illustrative embodiments, this description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative embodiments, as well as other embodiments of the invention, will be apparent to persons skilled in the art upon reference to the description. It is therefore intended that the appended claims encompass any such modifications or embodiments.

Claims

1. A method of operating a memory circuit, the memory circuit comprising a memory array having a memory portion and a spare portion, the method comprising: receiving a first write command to a

first memory address, wherein the first memory address has a status of being mapped to a first spare memory address, and wherein the first memory address corresponds to a first memory location in the memory portion and the first spare memory address corresponds to a first spare memory location in the spare portion; performing, in response to the first write command, a first write operation by attempting to write first data to the first memory location; determining if the first write operation is successful; and unmapping, in response to the first write operation of being successful, the first memory address from the first spare memory address.

2. The method of claim 1, wherein the first memory address having the status of being mapped to the first spare memory address indicates that the first memory location is considered failed memory bits.

3. The method of claim 1, subsequent to the first spare memory address being unmapped from the first memory address, further comprising: mapping a second memory address to the first spare memory address, wherein a second memory location associated with the second memory address is considered failed bits.

4. The method of claim 1, wherein unmapping the first memory address from the first spare memory address comprises disassociating the first memory address from the first spare memory address in a lookup table.

5. The method of claim 1, further comprising: performing a second write operation by attempting to write second data to the first memory location when the first memory address does not have a status of being mapped to a spare memory address; and in response to a determination that the second data was not successfully written to the first memory location: mapping the first memory address to a second spare memory address, wherein the second spare memory address is not mapped to any memory address, and writing the second data to a second spare memory location in the spare portion, the second spare memory location corresponding with the second spare memory address.

6. The method of claim 5, wherein the first spare memory location is the same as the second spare memory location.

7. The method of claim 5, wherein mapping the first memory address to the second spare memory address comprises associating the first memory address with the second spare memory address in a lookup table.

8. The method of claim 1, further comprising, before performing the first write operation, reading data for the first memory address from the first spare memory location.

9. The method of claim 1, wherein determining if the first write operation is successful further comprising: performing at least one verification read operation to verify if the first data is written to the first memory location successfully.

10. The method of claim 1, wherein the memory portion is physically separated from the spare portion in the memory array.

11. A memory circuit, comprising: a memory array comprising a memory portion and a spare portion; and a controller, configured to: perform a first write operation by attempting to write first data to a first memory location in the memory portion, wherein a first memory address corresponds with the first memory location, and the first memory address has a status of being mapped to a first spare memory address corresponding with a first spare memory location in the spare portion when performing the first write operation, and in response to a determination that the first data was successfully written to the first memory location during the first write operation, unmap the first memory address from the first spare memory address.

12. The memory circuit of claim 11, wherein the memory portion of the memory array and the spare portion of the memory array comprise resistive random access memory (RRAM) cells.

13. The memory circuit of claim 11, wherein the controller is configured to unmap the first memory address from the first spare memory address by disassociating the first memory address from the first spare memory address in a lookup table.

14. The memory circuit of claim 11, wherein the controller is configured to: perform a second write operation by attempting to write second data to the first memory location when the first memory address does not have a status of being mapped to a spare memory address; and in response to a determination that the second data was not successfully written to the first memory location: write the second data to a second spare memory location in the spare portion, the second spare memory location corresponding with a second spare memory address, and map the first memory address to the second spare memory address.
15. The memory circuit of claim 14, wherein the controller is configured to map the first memory address to the second spare memory address by associating the first memory address with the second spare memory address in a lookup table.
16. The memory circuit of claim 11, wherein the controller is configured to, before the first write operation is performed, read data for the first memory address from the first spare memory location.
17. The memory circuit of claim 16, wherein the controller is configured to, after the first memory address is unmapped from the first spare memory address, read data for the first memory address from the first memory location.
18. A system, comprising: an external circuit configured to generate instructions; and a memory circuit configured to receive the instructions from the external circuit, the memory circuit comprising: a memory array comprising a memory portion and a spare portion, and a controller, configured to, in response to a first instruction from the external circuit: perform a first write operation by attempting to write first data to a first memory location in the memory portion, wherein a first memory address corresponds with the first memory location, and the first memory address has a status of being mapped to a first spare memory address corresponding with a first spare memory location in the spare portion when performing the first write operation, and in response to a determination that the first data was successfully written to the first memory location during the first write operation, unmap the first memory address from the first spare memory address.
19. The system of claim 18, wherein the controller is configured to unmap the first memory address from the first spare memory address by disassociating the first memory address from the first spare memory address in a lookup table.
20. The system of claim 18, wherein the controller is configured to: in response to a second instruction from the external circuit: perform a second write operation by attempting to write second data to the first memory location when the first memory address does not have a status of being mapped to a spare memory address; and in response to determining that the second data was not successfully written to the first memory location: map the first memory address to a second spare memory address, and write the second data to a second spare memory location in the spare portion, the second spare memory location corresponding with a second spare memory address.
21. The system of claim 18, wherein the controller is configured to: before the first write operation is performed, read data for the first memory address from the first spare memory location; and after the first memory address is unmapped from the first spare memory address, read data for the first memory address from the first memory location.